

SHIVAM ARORA

✉ arshiv@umich.edu | 🌐 Shivam | 📄 arshiv-web.github.io | 📞 Shivam | 📱 +1-206-593-9817

EDUCATION

University of Michigan

M.S. in Computer Science & Engineering

Ann Arbor, MI

Aug 2025 – Present

Planned Coursework: Human-AI Interaction, Probability & Random Processes, Applied GPU Programming, Algorithms, Advanced Database Systems, Machine Learning, Advanced Scalable Systems, Natural Language Processing, Reinforcement Learning Theory

Thapar Institute of Engineering and Technology

B.E. in Computer Science; GPA: 3.98/4.00 (9.77/10.00)

Punjab, India

Aug 2017 – Jun 2021

TECHNICAL SKILLS

Programming: Python (NumPy, Pandas, OpenCV, Seaborn, Matplotlib); Java (Spring, DropWizard);

C/C++; Javascript; CUDA

ML & MLOps: PyTorch; Scikit-learn; Transfer Learning; MLflow; Gradio; GPU; HPC; Slurm; Pickle

Data & Cloud: Apache Spark; Snowflake; AWS (Glue, Lambda, S3, Secrets Manager); Terraform; DB2; HDFS; REST APIs

SWE Tooling & Platforms: Git; JUnit; Unix/Bash; React; Swagger; Reladomo; Legend (GS); Sybase IQ; Arduino; Verilog

WORK EXPERIENCE

Goldman Sachs

Associate - Technology

Bengaluru, India

Jan 2024 – July 2025

- Led architecture of a **Spark, AWS Glue, and Snowflake-based** insurance calculator, **processing 6M+ daily deposit accounts with a 20-minute SLA**. Delivered accurate pre-9AM ET calculations enabling timely deposit record keeping.
- Architected one of the **firm's first cloud-native AWS data pipelines** to process US deposit information, including **customer PII**. This project **improved data precision by 99%**, drove data standardization between divisions and ensured compliance with federal regulations (**FDIC Part 370**)

Goldman Sachs

Analyst - Technology

Bengaluru, India

Jul 2021 – Dec 2023

- Automated and streamlined** end-to-end German Deposit Protection and other European regulatory reporting systems using **Java, SQL, REST APIs, and Sybase IQ**.
- This effort **improved the EdB regulatory rating from E (Fail) to B and reduced quarterly reconciliation effort by 60%**. Received the **GS Controllars India Award (2022)** for exceptional performance in architecting this solution.

Goldman Sachs

Intern - Technology

Bengaluru, India

Jan 2021 – Jun 2021

- Developed a full-stack reconciliation platform (**React, Java Spring**) that automated reconciliations between firms General Ledger and Regulatory Reporting data, **serving 500+ professionals (Controllars) and saving over 800 person-hours each quarter**. This performance earned a **Pre-Placement Offer**. Collaborated in an **Agile environment, transitioning 300+ legacy screens to modern GS Web stack with comprehensive JUnit testing framework**.

SELECTED RESEARCH & PROJECTS

Eye Disease Classification 🦋

- Fine-tuned a **ResNet-18** with differential learning rates to classify eye diseases; enforced aspect-ratio-preserving preprocessing and training-only normalization to prevent data leakage.
- Achieved **95.26% accuracy, 93.82% macro F1, and 94.17% macro recall** on a held-out study dataset.
- Ensured **reproducibility and UX**: MLflow for experiment traceability; Gradio inference UI with accept/reject feedback loop and softmax confidences.

Brain Signal Controlled Prosthetics

- 2-Month Research Internship**, Indian Institute of Technology (IIT), Delhi; **Advisor:** [Dr. Soutik Betal](#)
- Developed an EEG-based prosthetic control system that integrated hardware with a **Convolutional Neural Network (CNN)**, **reducing setup time by 75% and enabling high-accuracy**, real-time prosthetic control.

Navigation System for the Visually Impaired 🦋

- Architected an IoT wearable using Arduino/Node.js + Google Maps; 99.2% obstacle detection with sub-3 second haptic response. Developed @ Thapar Institute of Eng. & Technology, India; **Advisor:** [Dr. Ashutosh Mishra](#)